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AI-Based Answer Script Evaluation Using OCR, Semantic Analysis and Generative AI

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Abstract—To evaluate descriptive answers, it is necessary to understand student's answer, expected concept and associated marking scheme. If documents are scanned or handwritten it is harder to process information from images and convert it to usable text. We present AI Answer Evaluator, a prototype solution for supporting teachers' assessments that combines optical character recognition, semantic analysis and generative artificial intelligence in a teacher-facing application. The system architecture consists of text extraction using PaddleOCR, reference context retrieval using Qdrant, and language model evaluation based on rubrics coordinated by a FastAPI layer. The workflow is exam preparation, extraction of question paper, review of students' answers, provisional scoring, feedback creation and final approval by the teacher. The marks allocated and awarded to criteria are presented in the user interface so that the teacher can review deductions alongside the student's answer. As per the operating systems case shown, 4 question marks of 8/10 equals 32/40 in total, as also seen in the teacher's decision panel. The results show the application workflow and the concerns about the reference content and mark allocations that still need to be verified. No metrics of the OCR accuracy or of agreement to an independent human grading were achieved at the time. The system is a useful platform for combining processing of answer scripts and contextual evaluation without taking away the final decision from the teacher.

Keywords—Semantic Analysis, OCR, Answer Script Evaluation, Generative AI, NLP, Automated Assessment.

I. Introduction

The answer provided by the students in a descriptive test cannot be evaluated on the basis of string matching algorithm as there is a chance that students might have used an alternative vocabulary, partial correctness or missed out an essential intermediate step. The evaluator should therefore consider the context of the marking scheme and the test question. The content of the handwritten work should be interpreted in terms of format, symbols, tables and clarity apart from the content. The issues of semantic and structural similarity between student answers and reference solutions in the automated assessment of short answers have been discussed in the literature [1].

Lexical and semantic properties are also used by modern algorithms to score the answer [2]. While there are reasons to use AI in assessment, the more efficient workflow is document processing, aligning questions and answers, justifiable reasoning and decision making. The AI Answer Evaluator takes care of this. It takes a digitized document, processes it, pulls relevant content from academic sources and calculates marks and feedback according to the marking schemes. The teacher can view the student answer, reference material, initial marks and final decision.

This paper describes the processing architecture consisting of three modules and application screens to handle examinations, subjects, question papers and scripts. The paper presents the processing architecture, the implementation workflow, and a demonstration of the functionalities of the prototype. Based on this study, the accompanying survey, the application architecture and screenshots, the system is described in the paper. The runtime configuration and performance benchmarks for the experiments were not set by the researcher, so results should not be considered as a true performance demonstration.

II. Related Work

A. OCR-Based Handwritten Text Recognition

Du et al. [3] proposed PP-OCR, an efficient OCR system that includes text detection, direction classification, and recognition. The modularity of the pipeline is pertinent to the objective of the current project to transform scanned scripts to text. The model may have a broad recognition ability but cannot guarantee the accuracy on a specific test set due to the variety of handwriting, mathematical expressions and page layout. Li et al. [4] proposed TrOCR, a method that utilizes image and text Transformers to recognize text in an end-to-end manner after pretraining and task-specific fine-tuning. The approach is promising due to the influence of its results on the development of methods for recognition of handwritten text. Recognition, however, is a separate task and is very different from the necessity to decide whether to award marks for an answer.

The text-extraction component in this project is documented as the PaddleOCR. The TrOCR is mentioned for the use as related work, not implemented as a dependency.

B. Automated Answer Evaluation

Mohler et al. [1] used a combination of semantic similarity and dependency-graph alignment to model short-answer grades. The method does not focus on counting the common words between the student answer and a reference solution. But the use-case of the paper is limited to short answer questions. The presented project is also dealing with scanned page processing and question segmentation.

The GradeAid framework developed by del Gobbo et al. combines lexicon and semantic features of answers with regression methods, with the evaluation on several datasets [2]. Such methods are relevant to the project due to the need for comparison and question-specific validation. The current project extends the document extraction and teacher review step around the evaluation, thus, the grading scope is concerned, but the higher accuracy is not claimed.

C. Semantic Similarity and NLP-Based Evaluation

Reimers and Gurevych propose Sentence-BERT to obtain sentence embeddings that are suitable for downstream tasks, including semantic similarity [5]. The method allows retrieving the relevant documents when the expected concepts are stated with paraphrases.

However, semantic similarity does not equal correctness: two answers may describe the same general idea but differ in a crucial detail. Lewis et al. proposed retrieval augmented generation which comprises a generative language model and an external retrieval component [6]. The point of interest is the use of a retrieval system to provide supporting evidence at inference time. In the context of assessment this supporting evidence is valuable only if the retrieved passage is related to the question and marking rubric. A high retrieval score or fluency of the generated response are inadequate proof that a mark is deserved.

D. Generative AI for Educational Assessment

Simultaneous essay scoring and feedback generation was investigated by Stahl et al. [7]. Their results show that scoring and feedback helpfulness are dissociated and therefore should be evaluated separately. Further, the paper is concerned with essay scoring, not the current task of structured operating-systems questions. Hashemi et al. [8] propose a multidimensional rubric scoring system and a learned calibration model in LLM-Rubric. The paper is concerned with scoring of dialogs, but the findings are also relevant to marking of rubrics and calibration against human scores. This project employs rubric-based evaluation as a feature of the design, not as a replication of the LLM-Rubric trained calibration process.

E. Research Gap

The areas of research that have been reviewed offer interesting building blocks but do not attempt the engineering challenge of bringing them together into an inspectable, answer-script pipeline. For such a task it is necessary for errors during extraction to be traceable back to the original page, that retrieved passages relate to the correct question and individual marks are understandable against the rubric. It therefore retains the processing steps joined together and facilitates teacher review. Its contribution is in terms of application level integration, not a different OCR architecture or experimentally established improvement over the cited grading methods.

III. Proposed System

A. System Overview

The system takes into account the context of examination, question paper, student scripts and reference material. The processing flow involves upload, validation, OCR, answer segmentation, semantic retrieval, evaluation, verification, feedback and outputs. The general architecture of the project is depicted in Figure 1, and its processing pipeline is summarised in the form of three technical modules, which are explained below. The higher level application also contains a set of management, service and storage functionalities.

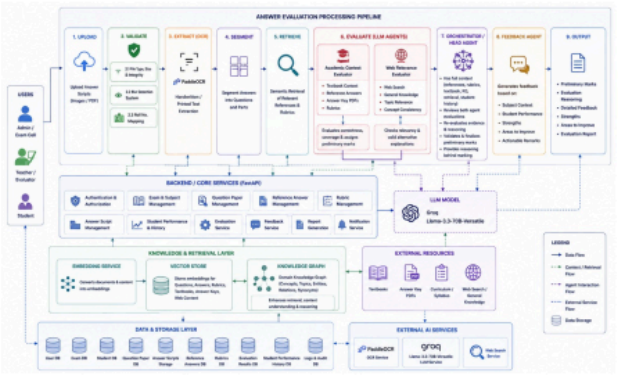


Fig. 1. Overall system architecture. The processing pipeline, core services, retrieval layer, and evaluation roles are illustrated. The three paper-level modules group these responsibilities.

The architecture contains FastAPI core services placed between the external AI components and the user interface.

PaddleOCR is used for the text extraction process whereas the retrieval is performed by Qdrant. The evaluation stack is made up of a Groq-hosted language model, academic and web relevance evaluators, an orchestrator and a feedback agent. The process of examination administration, reference answers, rubrics and results storage facilitates the process flow. The architectural design consists of knowledge-graph enrichment, live web retrieval, and notifications, which are not used in the functionality that is presented.

B. Module 1: Answer Script Preprocessing and OCR

The first module involves the conversion of visuals into text, which is related to a particular question. In relation to the design, the process consists of the validation of files, which ensures the integrity of the documents and prepares them for recognition.

The architecture contains the blur detection and roll number mapping. The interface presents a page view with a clarity score. This enables the viewer to preview the document and verify that the extracted text is of acceptable quality. The available material does not specify particular algorithms for de-noising, de-skewing or thresholding. The OCR results must be structured enough to distinguish between question numbers, the body text, table sections and answer sections. The next step is to map text to the corresponding question and sub-part.

This operation is crucial as a correct recognition but misplaced association will lead to wrong scores.

The raw and clean views allow the reviewer to evaluate the extracted content before accepting the evaluation.

C. Module 2: Semantic Answer Analysis

The second module performs the link between the student answer and the expected concepts by extracting the reference information to support it. Let q_i be a question, a_i its extracted answer, R_i its rubric and D the reference collection. The retrieval formulation based on embeddings is

$$C_i = \mathrm{TopK}_{d \in \mathcal{D}} \, \mathrm{sim}\big(f(q_i), f(d)\big), \tag{1}$$

where f is a text-to-vector mapping, and C_i is the retrieved context. One possible similarity measure is

$$\mathrm{sim}(\mathbf{u}, \mathbf{v}) = \frac{\mathbf{u}^{\mathsf{T}}\mathbf{v}}{\|\mathbf{u}\|_2 \|\mathbf{v}\|_2}. \tag{2}$$

Equations (1) and (2) define the retrieval design. Qdrant offers vector search and payload filtering [9] which allow the subject and question metadata to restrict the reference context. The retrieved material is used to inform the rubric-based decision but is not substituted. Then the student answer is assessed with the retrieved reference and rubric. The answer could be semantically related but incomplete, or use correct terminology in a wrong conclusion. Hence, similarity is used for context selection and conceptual comparison, not directly converted to marks assuming an assumed threshold or weighting formula.

D. Module 3: Generative AI-Based Evaluation and Feedback

The third module uses the question, extracted response, reference context, marking criteria and asks for structured evaluation. In the architecture, the academic evaluator considers correctness and coverage, and the web relevance evaluator is supposed to assess related or alternative explanations. The orchestrator evaluates them before giving preliminary marks. The feedback stage describes strengths and areas that need improvement. This role-based organization distinguishes academic assessment, consistency review and feedback generation. Its output has criterion-level marks, succinct evaluation reasons and advice a teacher can inspect.

The teacher's decision comes as the last chance to take charge as the teacher decision is the final record while the AI scores are explicitly preliminary.

E. Evaluation Workflow

The evaluation workflow includes the following steps:

1) Create / Select exam & subject. Attach question paper and marking scheme.
2) Upload the document and review the extraction and question structure at the page level.
3) Link a student script to the examination, identify text and match answers to questions.
4) Collect appropriate reference materials and score each response with its rubric.
5) Provide preliminary scores, criterion level explanations and student feedback for review.
6) Let teacher see mark or change and save final decision. Summarise script status in analytics.

The application describes these phases through management, extraction, evaluation and approval screens.

IV. Implementation

A. Input and Answer Script Processing

The application categorises the submissions by exam and subject. The teacher starts the process by creating an examination. The subjects of the examination are included and the question paper and marking scheme are connected. The subject-management screen is shown in figure 2. It has controls to upload a question paper, view the extracted questions from it, and upload an answer script.

The association supplies the required scholarly resources to accommodate the interpretation of any response.

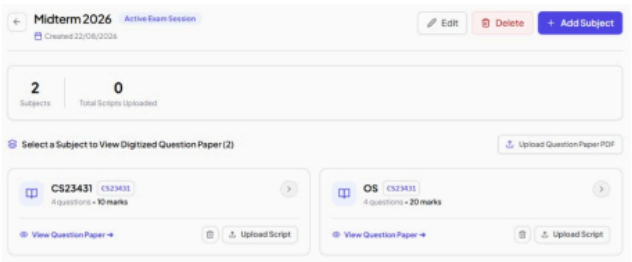


Fig. 2. Question-paper and answer-script upload controls for the exam and subject setup.

The session has twelve pages for student script and two pages for question paper. The document processing framework uses the pages as scanned pages or images, performs a validation and preps it for OCR. The design considers student identity recognition, blur detection and file integrity. The clarity tool and navigation tools of the page viewer provide a content and quality assessment of the document before assessment. Table I organizes the component responsibilities.

The Python backend controls the various processes, while the frontend is the user interface to input documents and see the results of the assessment. Reproducibility requires a source-level configuration record of framework versions, storage schemas and inference settings.

B. OCR Implementation

PaddleOCR is the text extraction module of the project, which has a core processing file called `paddle_ocr.py`. The interface of the module consists of two windows: the scanned page and the extracted question cards. Each question card has the text of the question, marks, table data in structure if it was detected, and an edit control. The figure below shows the first question from the operating systems paper with extracted text.

In addition, the system has a review flag to approve results before using them. Moreover, numbers and boundaries of the questions are associated with the assessment items to ensure

TABLE I
APPLICATION COMPONENTS AND THEIR RESPONSIBILITIES

Component	Responsibility
Frontend	Examination setup, extraction review, marks, feedback and approval
FastAPI backend	Control document processing and evaluation services
PaddleOCR	Recognition of text in uploaded pages
Qdrant / RAG	Identification of relevant reference material for assessment
Groq / Llama	Assessment according to a rubric and receiving feedback
Teacher review of results	Validation or adjustment of initial scores prior to finalisation

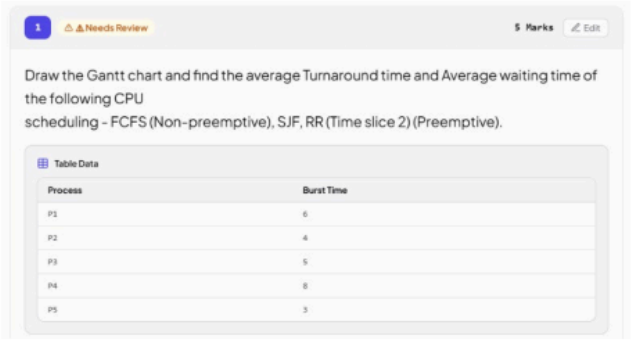


Fig. 3. Extracted question card with structured process and burst-time data. The review flag allows the result to be checked before use.

the correct correspondence between extracted text and the items.

The interface keeps scheduling parameters in the separate columns of the table, allowing to look at values that would not be intuitive to put together in one continuous string. The question cards, marked "Needs Review," recommend for the contents extracted to be reviewed. Student responses also have the raw and clean text views to be able to look at the artifacts of recognition next to the produced evaluation.

C. Semantic Analysis Implementation

The semantic-analysis process combines the question, student response, and reference context. The reference material is organized in a searchable vector collection, and Qdrant supplies semantically relevant content to the evaluator.

This reflects the companion survey's distinction between semantic relatedness and correctness: an answer can be correct on the topic, but miss a required notion or apply a formula wrongly. The review interface shows student answer and reference content in parallel, and the reviewer has to understand both in the context of the marking scheme's correctness, relevance, coverage, and completeness. The current screens do not show the retrieved passage identifiers or the numerical similarity scores, and thus the quality of retrieval is reflected by the design as documented, and quality of retrieval will be deferred to later testing.

D. Generative AI Evaluation Implementation

The architecture identifies Groq and `llama-3.3-70b-versatile` for language-model evaluation. The documentation reports that the model accepts and outputs text with JSON-object support [10]. Hence, the OCR provides the input text for the evaluation process. The model ID is fetched from the architecture while the deployed configuration needs to be verified from the inference logs. The evaluation request consists of the question, recognized answer, reference evidence, marks, and rubric. The output provides the question-wise marks, criterion scores, explanation, and feedback. The Figure 4 shows the evaluation summary along with the Academic Evaluator, Web Relevance Evaluator, and Orchestrator (Final) panels.

These panels organize the review by responsibility; the interface alone does not establish individual model calls for each role.

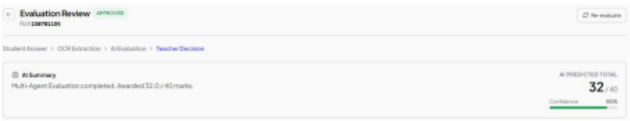


Fig. 4. Evaluation review summary with an AI prediction of 32/40 and approved review status.

E. Score and Feedback Generation

Marks are shown in both criterion and question levels. For question i let w_{ij} be the marks allocated for criterion j, r_{ij} be the marks awarded and M_i the question maximum. The aggregation is stated as

$$s_i = \sum_{j=1}^{m_i} r_{ij}, \qquad M_i = \sum_{j=1}^{m_i} w_{ij}, \qquad 0 \le r_{ij} \le w_{ij}. \tag{3}$$

The preliminary script total is $S_{\mathrm{AI}} = \sum_i s_i$.

Figure 5 shows the displayed rubric breakdown and explanation of deduction. The separated allocated and earned values draw attention to the partial credit for the reviewer.

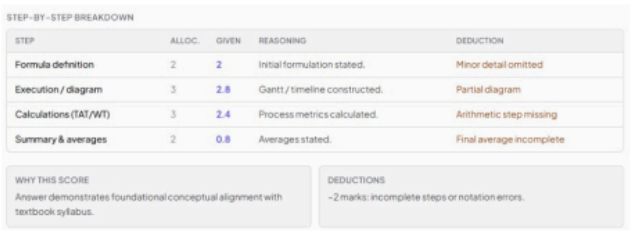


Fig. 5. Criterion-level mark allocation and deduction explanation for the selected question.

Feedback improves the numerical outcome by highlighting strengths and pointing out areas for improvement. In Figure 6 the advice tells the student that they need to enhance the statements of the formulae and the intermediate working.

Correlating feedback with the rubric enables the student to see the section of the answer that needs their attention.

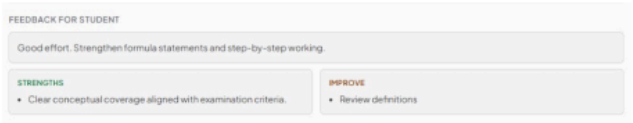


Fig. 6. Student feedback identifying advice on improvement and a summary of strengths.

The teacher decision-screen consists of editable question marks, the final total, optional notes, and "Approve & Save." The AI marks are tentative suggestions from the AI for the teacher to edit and finalize their grades and total (the summation of $\sum_i h_i$). The methodology will permit a teacher to make their decision vis-à-vis the recommendation of the AI.

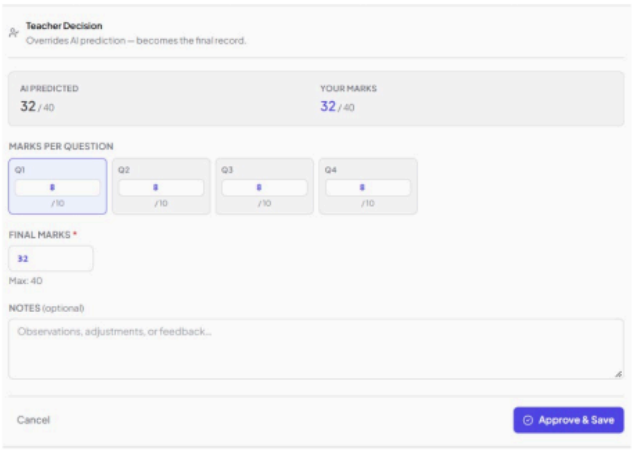


Fig. 7. Teacher review interface with editable question marks, final total, notes, and approval control.

F. User Interface / Evaluation Dashboard

The interface comprises the dashboard on which the teacher can view the exams, scripts, AI's evaluations, and awaiting reviews.

Its four stages of the process are Examinations, Question Paper, Upload Scripts, and Review & Approve. Figure 8 shows the dashboard overview.

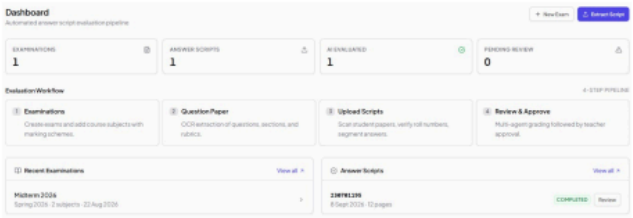


Fig. 8. Dashboard overview with processing counts, the four-stage process, and the completed-script entry.

The review page also splits the stage into Student Answer, OCR Extraction, AI Evaluation, and Teacher Decision and allows the teacher to see the supporting information before finalizing on the result. The analytics page gives a summary of the status of the process over examinations. Figure 9 shows the demonstration session with one evaluated script and no pending review.

This view of the status complements the question level assessment to indicate the progress of the uploaded scripts through the processing workflow.

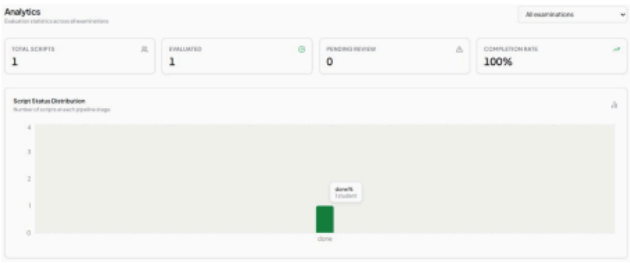


Fig. 9. Analytics view of the demonstration script in the complete processing state.

V. Results and Analysis

The demonstration is based on the operating systems question paper and a review session of four scored questions. The dashboard contains a reporting interface to an individual evaluated script. Subsequent analysis represents the output of the extraction, scoring, feedback and final review interface. Screens of different aspects of that session are not independent test cases (the study represents a functional demonstration rather than a controlled accuracy experiment).

A. OCR Performance

The extracted question contains the CPU-scheduling instruction and the tabular parameters. The five process/burst-time pairs, P1–6, P2–4, P3–5, P4–8, and P5–3, are all found in the original visible document. This represents preservation of the presented printed table, not a general handwriting-recognition result. The view of the student-answer is not so clear-cut: abbreviations and numeric fragments are recognizable, but neither the original handwritten answer nor a full manual transcription is available for comparison. The quantitative evaluation of the quality of the recognition process requires reference transcriptions. Character error rate (CER) is conventionally defined as

$$\mathrm{CER} = \frac{S_c + D_c + I_c}{N_c}, \tag{4}$$

In this expression, character substitutions, deletions, and insertions are denoted by S_c, D_c, and I_c, respectively, and N_c is the number of reference characters. While the word-error rate is based on a similar edit distance (applied to words), these measures are not reported (as there is not an aligned text). Additionally, the page-clarity value is not a measurement of the OCR accuracy.

B. Semantic Evaluation Performance

The semantic evaluation performance relates the output to CPU scheduling concepts, stressing the importance of getting the concepts right (aligned with the syllabus). A possible next test would be to ensure that different phrasing of the same answer gets the same credit, but incorrect calculations still count as errors.

This distinguishes our understanding of the answer from the recognition of the topic. No semantic similarity score or retrieval metric is available. In the case of the sample question, the panel labeled “Reference Answer” restates the question prompt, not a worked example; the actual reference evidence that the evaluator has been provided needs to be checked. We recommend that evaluation of this module separately measure the relevance of retrieved context and concept-level judgments. In particular, a scheduling response can discuss FCFS, turnaround time, and waiting time with an incorrect timeline: topic agreement is insufficient to accept such a response.

C. Generative AI Evaluation Results

Table II displays the numeric rubric breakdown for Question 1. The four values awarded add up to us giving us $2.0+2.8+2.4+0.8 = 8.0$ which is same as that of the question mark represented. Therefore, the total deductions have been of two marks. The system’s feedback towards the response is that they wish to see the student improve on statement of formulae and to show intermediate working; The response acknowledges coverage of concepts as being a strength.

TABLE II
DISPLAYED QUESTION 1 RUBRIC BREAKDOWN

Criterion	Allocated	Awarded	Displayed issue
Formula definition	2	2.0	Minor detail omitted
Execution / diagram	3	2.8	Partial diagram
Calculations (TAT/WT)	3	2.4	Arithmetic step missing
Summary & averages	2	0.8	Final average incomplete
Total	10	8.0	Two marks deducted

Figure 5 below shows the application screen. The format of explanations put forward has made it easier for a reviewer to identify the items deducted. On the contrary, the formula row has been given full marks whereas a deduction field indicates a detail has been missed out.

This would not necessarily prove the score wrong, but it shows that there is a problem with the statement of the score for the purposes of review. Generic reason like conceptual congruence is a much worse reason as a reason than invoking a specific calculation or sentence used by the student.

D. Overall Evaluation Results

Table III contains the values at the script level which have been presented during the review session. All the four questions have 8/10 and demonstrate the same level of confidence value which is 85%. Their sum is 32/40 which is equal to 80% of the one shown maximum. The review header is marked as "Approved" and the teacher panel shows 32/40.

The following is purely for the purpose of checking the arithmetic for the two screens, there is no pretense that there is agreement with human marks derived independently

TABLE III
OBSERVED VALUES IN THE CAPTURED REVIEW SESSION

Item	Displayed value	Interpretation
Question 1	8/10	Preliminary AI mark
Question 2	8/10	Preliminary AI mark
Question 3	8/10	Preliminary AI mark
Question 4	8/10	Preliminary AI mark
AI total	32/40	Sum of four marks
Teacher total	32/40	Value in decision panel
Confidence	85%	Calibration not verified
Analytics completion	100%	One of one scripts evaluated

Several limitations impinge upon interpretation.

First, the question-paper cards have five-mark questions, the subject card shows a twenty-mark paper and the evaluation screen shows four ten-mark questions and a forty-mark total — the captures do not establish whether these are different setups or a mapping problem, and therefore do not provide a fully reconciled trace of scoring. Second, the same marks and confidence values for all the questions could be a default response, a pattern, or an example; screenshots cannot show that it is not the former or the latter.

Finally, agreement between the AI and teacher panels is not an independent confirmation if the teacher values were copied or accepted from the prediction. The demonstration supports the availability of document review, structured evaluation, feedback and a teacher decision interface. It does not show the gain from processing time, versatility of handwriting, calibration of confidence, or grading accuracy. None of the numbers reported can have character error rate, mean absolute score error, correlation, or quadratic weighted kappa calculated from them.

E. Comparative Analysis

Table IV compares different approaches based on mechanisms and limitations.

It is a comparison of approaches on the qualitative level, not an empirical evaluation. The proposed workflow combines OCR, reference retrieval, rubric outputs and teacher review in one application which have yet not been directly compared to alternatives on the same set of scripts. An apples-to-apples comparison would test similar questions, marking schemes and independent human grading to allow for comparison approaches on these dimensions. Possible baselines include keyword matching, embedding-based grading, an LLM without retrieval, and the complete proposed pipeline.

The scores need to be calibrated on a separate development set, not fitted on the test answers. Comparing manually

TABLE IV
QUALITATIVE COMPARISON OF EVALUATION APPROACHES

Approach	Capability and limitation
Keyword matching	Simple and transparent, but poor handling of paraphrases, negation, and reasoning traces.
Semantic grading [1], [2]	Uses semantic or structural features; document extraction and review processes need separate integration.
Embeddings [5]	Efficient semantic matching, but similarity alone does not guarantee correctness or a calibrated score.
LLM rubrics and feedback [8], [7]	Criterion-oriented evaluation with explanations; requires human calibration and feedback validation.
Proposed workflow	Includes OCR, retrieval, scoring, feedback, and teacher review; controlled performance testing is needed.

transcribed answers with answers obtained with OCR would additionally isolate the effect of recognition errors.

For N responses, the grading errors can be summarized by $\mathrm{MAE} = N^{-1} \sum_{i=1}^{N} |s_i - y_i|$, where y_i is an independent human reference score. For different maxima, a common normalized scale should be adopted or scores reported per question. Human-human agreement should be established before interpreting human-AI agreement, and feedback evaluated independently for correctness, relevance and usefulness. These are all evaluation procedures for further experiments; there was no such comparison in the provided results.

VI. CONCLUSION AND FUTURE WORK

AI Answer Evaluator performs document extraction, semantic reference retrieval, generative evaluation, and teacher review as a proposed assessment workflow.

The prototype interface presents question-paper structuring, the question-wise marks, rubric breakdowns, feedback, and approval controls. In the given example, the score obtained is 32/40, which matches with the total in the teacher decision panel, for the four questions. This result is merely an example of the use of the review interface and not a proof of grading reliability. The analysis highlights specific areas of improvement, which have been prioritized for the implementation. These include the reconciliation of the question-paper and evaluation mark scales, providing worked reference answers together with traceable retrieved evidence and alignment of the deduction text with the criterion marks.

The OCR execution, language-model parameters, retrieval settings, and run-time traces should be logged prior to reporting results. A larger sample of independently graded, transcribed student work would be needed to evaluate recognition accuracy and agreement. More effort would be directed toward improving recognition of handwritten mathematical expressions and diagrams, evaluating in more than one language, and nuanced feedback per question. Confidence estimates must be calibrated against observed errors, and uncertain or inconsistent responses prioritized for teacher review. Retrieval ablations and reruns of scoring will be executed to evaluate whether additional components enhanced scoring and feedback to a degree deemed sufficient for adoption.

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